

3.04 Moments				
3.04a	Statics		a) Be able to calculate the moment of a force about an axis through a point in the plane of the body. <i>For coplanar forces, moments may be described as being about a point.</i> <i>[Understanding of the vector nature of moments is excluded.]</i>	MS1
3.04b			b) Understand that when a rigid body is in equilibrium the resultant moment is zero and the resultant force is zero.	
3.04c			c) Be able to use moments in simple static contexts. <i>e.g. To determine the forces acting on a horizontal beam or to determine the forces acting on a ladder resting on horizontal ground against a vertical wall.</i> <i>Questions will be set in which the context of the problem can be modelled using rectangular laminas, uniform and non-uniform rods only.</i> <i>Learners may assume that:</i> <i>1. for a uniform rod the weight acts at the midpoint of the rod,</i> <i>2. for a non-uniform rod the weight acts at either a specified given point or is to be determined by moments,</i> <i>3. for a rectangular lamina the weight acts at its point of symmetry.</i>	